Emily Louie

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EDUCATION

University of Waterloo – *Systems Design Engineering (Bachelors of Applied Science)* **2019 – 2024**

Courses: Data Structures and Algorithms (C++), Digital Systems (C) Cumulative GPA: 87% (3.9/4.0)

TECHNICAL SKILLS

Frontend: JavaScript/TypeScript, React, Redux, Gatsby, HTML, CSS/SCSS, GraphQL, REST

Backend: Java, Spring Boot, JUnit, C/C++, Golang, Python, Docker, Firebase, Azure, Neo4j, PostgreSQL, MongoDB

EXPERIENCE

Lumafield – *Software Engineering Intern* **Sep – Dec 2022**

- Iterated and developed a user-friendly interface for the world's first accessible x-ray CT scanner for engineers
- Architected and built an **authentication system** for the interface and API routes using **Python** and **Ansible**
- Optimized the re-rendering of the frontend to load **55% faster** using **TypeScript** and **React**
- Identified bugs in the firmware, and collaborated with other engineers to create a holistic solution in **C**

Figma – *Software Engineering Intern* **Jan – Apr 2022**

- Launched **3 new features** to enhance collaboration in FigJam which has 400,000+ weekly active users
- Consolidated numerous points of entry for users to increase productivity by **33%** using **TypeScript** and **React**
- Refined canvas snapping algorithms to improve user experience using **C++** with 100% testing coverage
- Built a system in TypeScript and React to display emoji reactions and custom CSS animations in real time for keywords typed in cursor chat for FigJam's first year anniversary, doubling usage of the chat feature

Unity – *Software Engineering Intern* **May – Aug 2021**

- Constructed and deployed **7+ advertising experiments** on Unity's ad network to 10+ billion users
- Generated **5% more revenue** by creating custom frontend layouts and animations in **TypeScript** and **React**
- Created a custom component library with 100% testing coverage that uses Storybook and Linaria

UW Blueprint – *Software Developer* **Aug 2020 – 2021**

- Created a full stack starter application with **MERN** and **Flask** for teams to develop software for nonprofits
- Designed a decision-making platform (arbitrium.web.app) that helped allocate **$2+ million** to local charities

Ontario Digital Service – *Software Engineering Intern* **Sep – Dec 2020**

- Developed and launched 3 new versions of Ontario's COVID-19 data visualizations (covid-19.ontario.ca/data) and informational microsites using **JavaScript** and **React**, averaging **450,000+ unique views** per month
- Reduced average build times by **52%** by optimizing data collection algorithms in the project's build process

Ontario Digital Service – *Product Design Intern* **Jan – Apr 2020**

- Collaborated with engineering teams to pioneer AODA/WCAG 2.0 compliant designs for Ontario's COVID-19 sites (covid-19.ontario.ca) which had **1.25+ million views** the month after launch

Microsoft – *Software Engineering and Product Design Intern* **Jul – Aug 2019**

- Launched and developed the frontend of an internal inventory tracker used by **2,000+ employees** and reduces the request processing time by over **85%**, using **JavaScript**, **React** and **Azure**

PROJECTS

Six Degrees of Spotify – *Java, Spring Boot, Neo4j, Docker, Vaadin, JUnit* **/six-degrees**

- Built a full stack web application to explore that artists are connected by six or fewer song collaborations by maintaining a Neo4j database with **450,000** collaborations parsed from the Spotify API